

Agenda

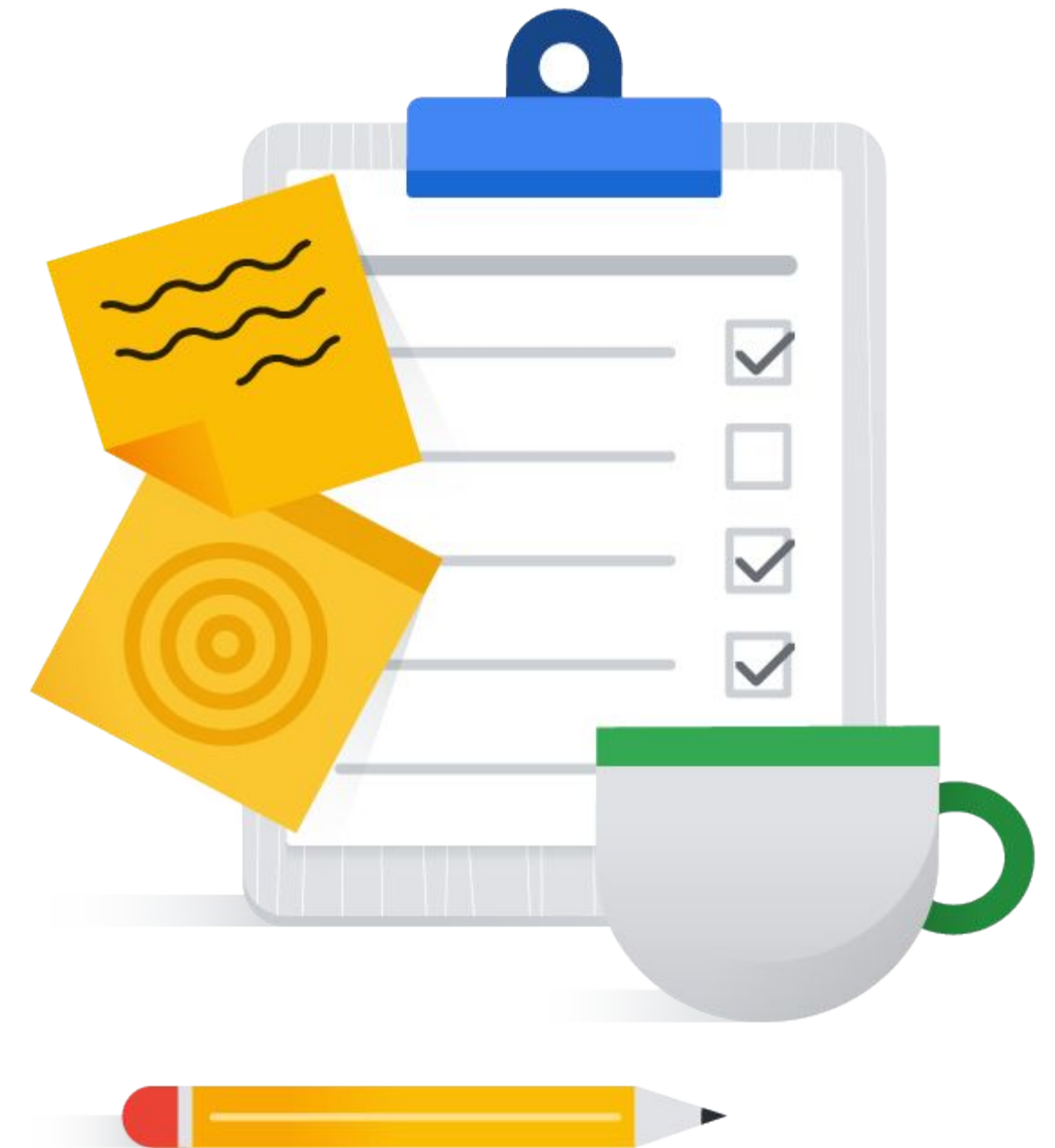
- 01 Introducing Agent Development Kit
- 02 Develop agents with ADK
- 03 Build multi-agent systems with ADK
- 04** Deploy ADK agents to Agent Engine
- 05** Evaluate ADK agent systems



Deploy ADK agents to Agent Engine

Module objectives

01	Describe Agent Engine's runtime environment.
02	Outline agent development and deployment using Agent Engine.
03	Identify the key benefits of using Agent Engine.

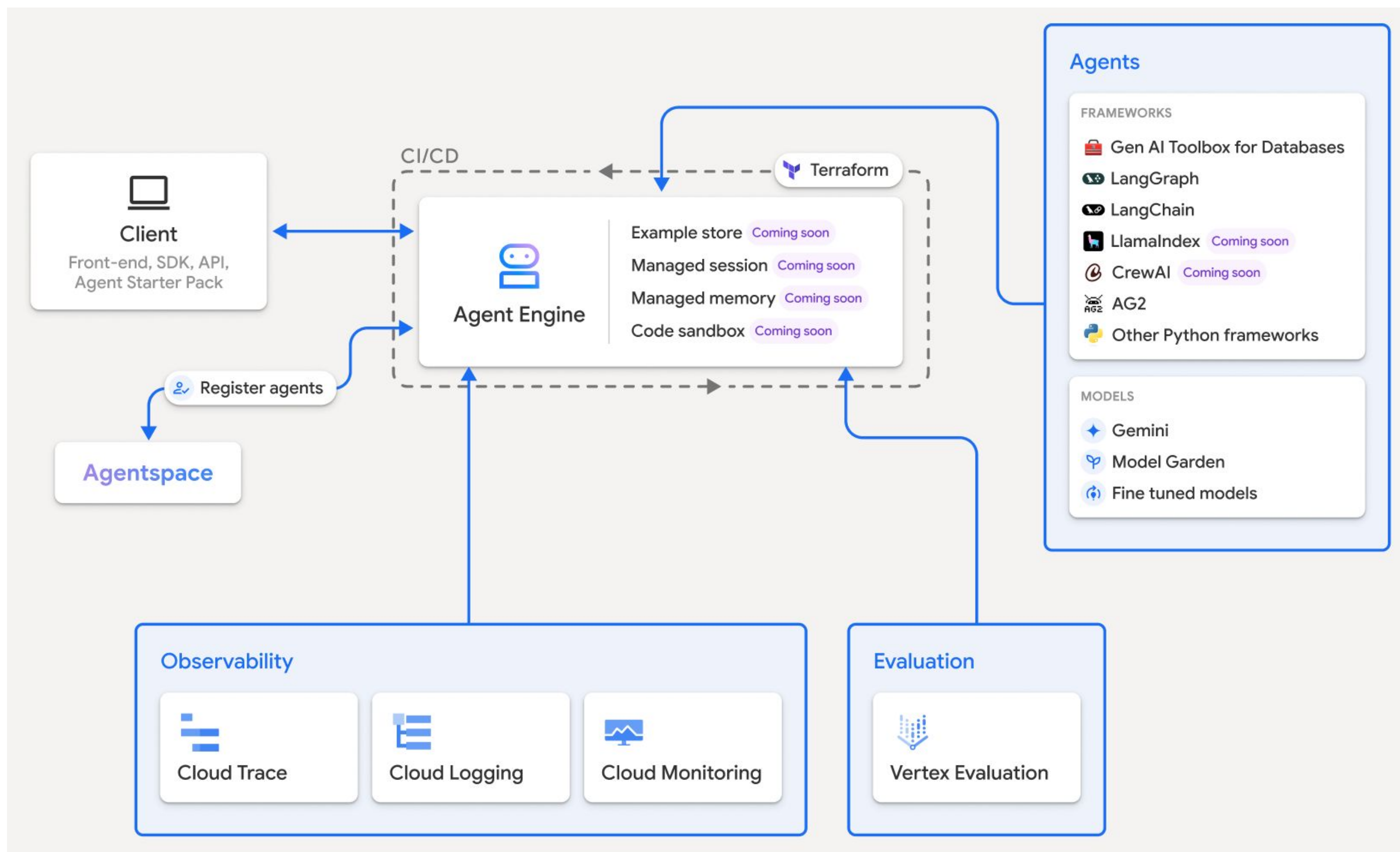


Agent Engine enables gen AI developers to develop, deploy, query, and manage gen AI apps using Agent Development Kit (or other open source frameworks) with secure, fully managed infrastructure on Vertex AI.

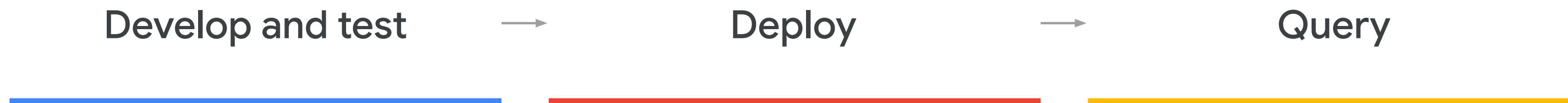
Agent Engine

Theme	Features
Scalable deployment	• Enables reliable and scalable agent deployments on Google Cloud • Provides an API to manage and query agents • Handles security (VPC-SC) and authentication concerns
Comprehensive agent management	• Offers tools for session and memory management • UI to list, view, and converse with agents • Facilitates agent monitoring and analytics
Quality and evaluation	• Use evaluation tools to measure and understand agent performance quality • Add to and manage the Example Store to improve agent performance
Framework agnostic	Supports deployments of agents built on various frameworks: • First-party Agent Framework from Vertex AI • Third-party open source frameworks such as LangGraph, LangChain, and Crew AI

Agent Engine: The keystone



Develop and deploy agents with Agent Engine



Develop and deploy agents with Agent Engine

Develop and test



Deploy



Query

```
model = "gemini-2.0-flash"

def get_exchange_rate():
    "Retrieves the exchange rate."
    ...

agent = reasoning_engines.LangchainAgent(
    model=model,
    tools=[get_exchange_rate],
    }
)
```


Develop and deploy agents with Agent Engine

Develop and test



Deploy



Query

```
remote_agent = agent_engines.create(  
    agent,  
    display_name="my_agent",  
    requirements=[  
        "google-cloud-aiplatform",  
        "langchain"])
```

Develop and deploy agents with Agent Engine

Develop and test



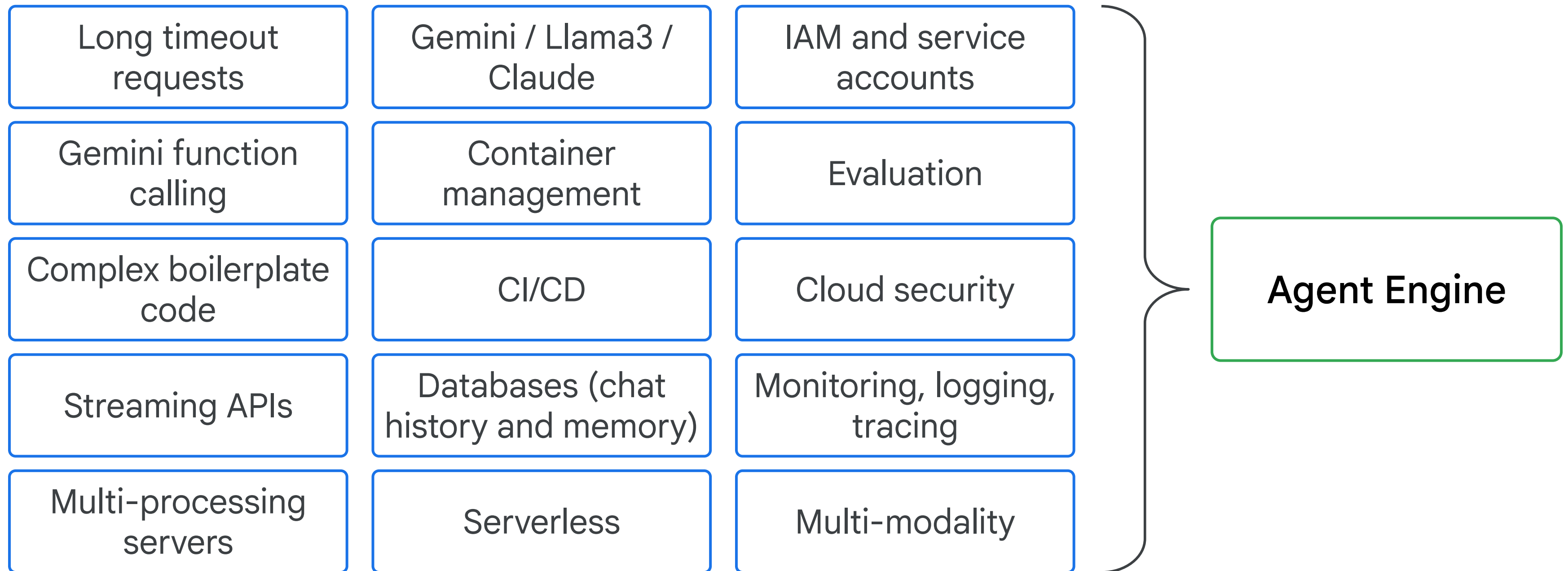
Deploy



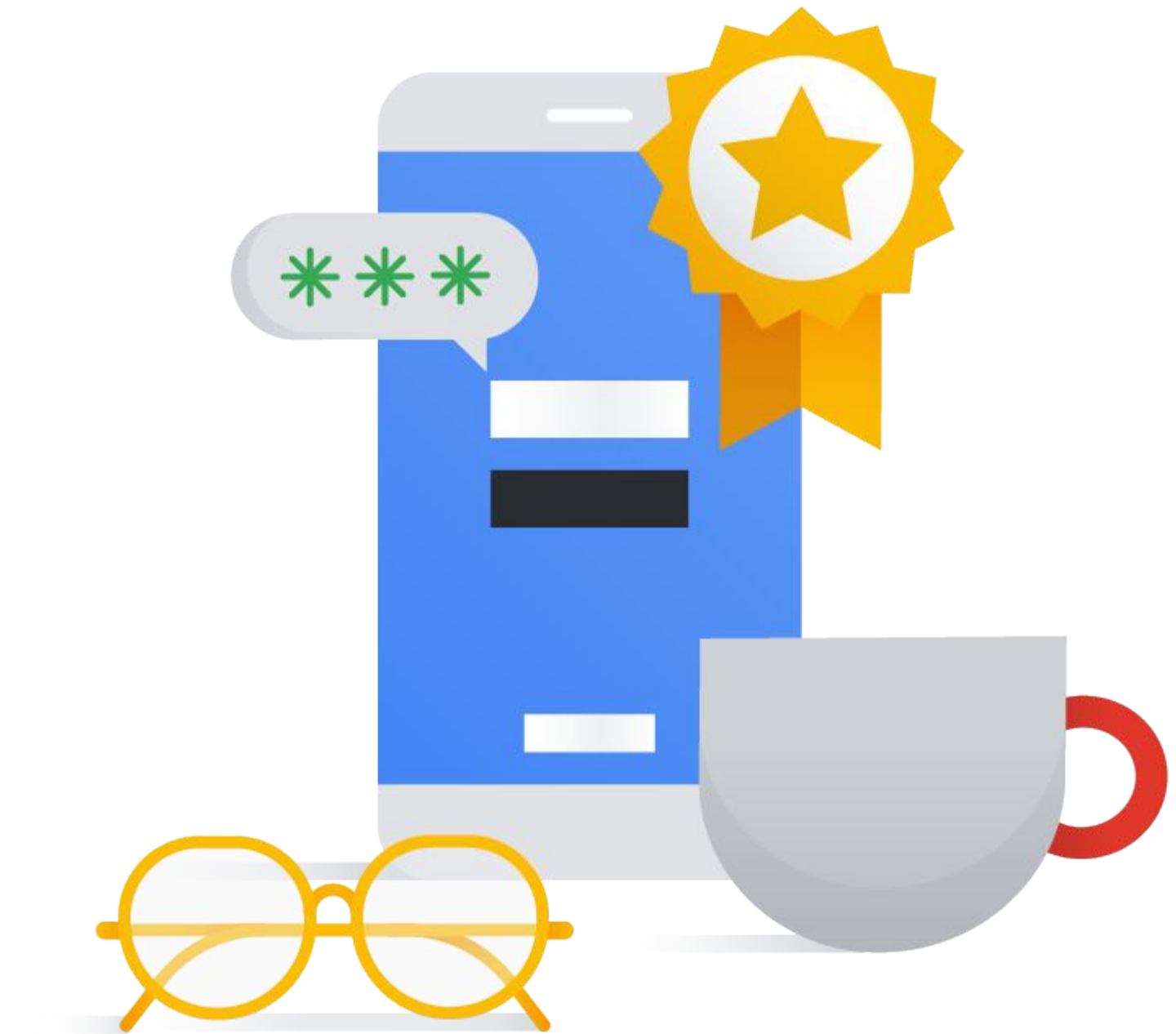
Query

```
remote_app.query(  
  query="What's the exchange rate from US  
dollars to Swedish currency?")
```

Agent Engine: The shortest path to success



Now let's do a short
quiz to **check your
knowledge!**



Quiz | Question 01

Question

What is Agent Engine primarily a runtime for?

- A. Deploying and scaling agents in production
- B. Developing new agent frameworks
- C. Creating user interfaces
- D. Testing individual models

Quiz | Question 01

Answer

What is Agent Engine primarily a runtime for?

- A. Deploying and scaling agents in production
- B. Developing new agent frameworks
- C. Creating user interfaces
- D. Testing individual models



Quiz | Question 02

Question

What level of infrastructure management does Agent Engine provide?

- A. Bare metal
- B. Fully managed
- C. Serverless
- D. Edge devices

Quiz | Question 02

Answer

What level of infrastructure management does Agent Engine provide?

- A. Bare metal
- B. Fully managed
- C. Serverless
- D. Edge devices



Quiz | Question 03

Question

What can you deploy to Agent Engine?

- A. Gemini models or ADK agents
- B. ADK agents, agents from other frameworks, and custom agents
- C. Only ADK agents
- D. ADK agents and CES conversational agents

Quiz | Question 03

Answer

What can you deploy to Agent Engine?

- A. Gemini models or ADK agents
- B. ADK agents, agents from other frameworks, and custom agents
- C. Only ADK agents
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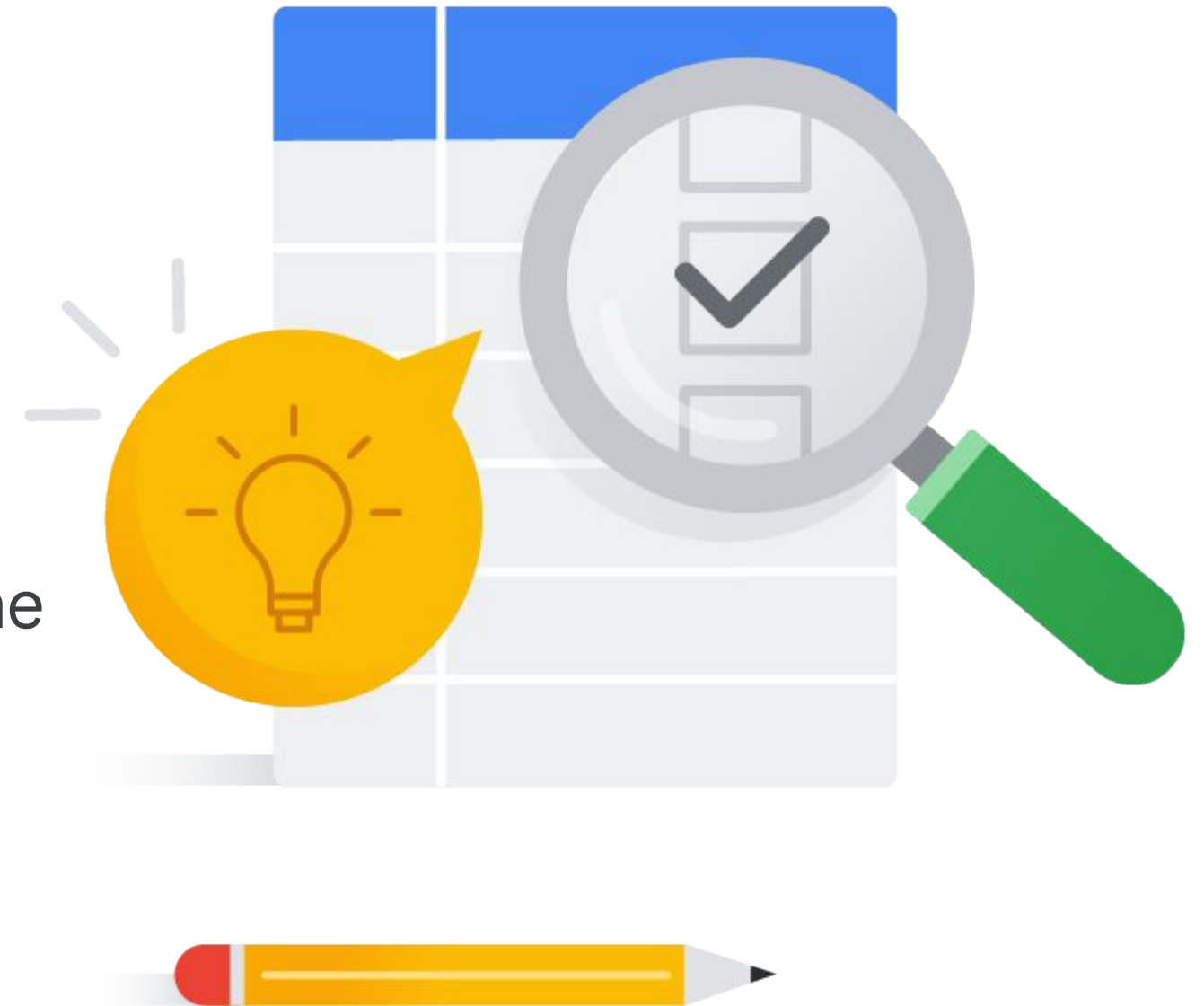


Lab: Deploy ADK agents to Agent Engine

🕒 45 min

Overview

In this lab, you will learn to deploy ADK agents to Agent Engine for a scalable, fully managed environment for your agentic workflows.



Google Cloud